

How To USE YOUR PHONE As A THERMAL CAMERA

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Thermal cameras, or thermal accessories, are useful for thermal scanning based fault detection, thermal throttle analysis, thermal monitoring, and many other purposes. Thermal scanning is also



Fig. 1: Author's prototype

used in monitoring the temperature of furnaces or boilers, or for analysing body temperature without physical contact—like detection of Covid-19 in a person.

While the thermal cameras are expensive with their price anywhere between ₹30,000 and ₹100,000, and sometimes even higher, some cameras are available as external accessories for mobile phones. These can convert a phone into a thermal camera, but even these accessories can cost up to ₹30,000, or higher, which many of us may find still

quite expensive for basic thermal analysis or thermal scanning or to test Covid-19 symptoms.

So, here is how you can make an affordable thermal camera accessory with which you can turn a phone into a thermal camera for as low as ₹5000 to ₹6000. This camera can be used to get a very basic thermal video and for other basic tasks like thermal screening, thermal testing analysis, and thermal monitoring.

(EFY note. An on-the-go adaptor (OTG adaptor) can be used according to the phone on which this device is to be configured. For iPhone, we will need an iPhone OTG adaptor, and for an Android phone, a type-C USB.)

First, the body of the device needs to be designed in a way that supports and protects the components and the device. The thermal sensor and the Raspberry Pi Zero W are used here together. The enclosure/case of the device can be designed as per the size of the Raspberry Pi Zero W. A hole can be made at the bottom of the case to expose the thermal sensor. A cut for the USB micro OTG outlet can be provided at the top of the case, as shown in Fig. 3.

After designing the case, connect the sensor, Raspberry Pi and OTG as shown in the circuit diagram in Fig. 4. If you have an OTG with a micro USB male output, you can directly connect that micro USB to the Raspberry Pi micro USB input. Or else, you need to cut the wire of the OTG and solder the +ve wire to the 5V pin of the Raspberry Pi, and GND to the GND pin of the Raspberry Pi. For the iPhone, avoid using the soldering

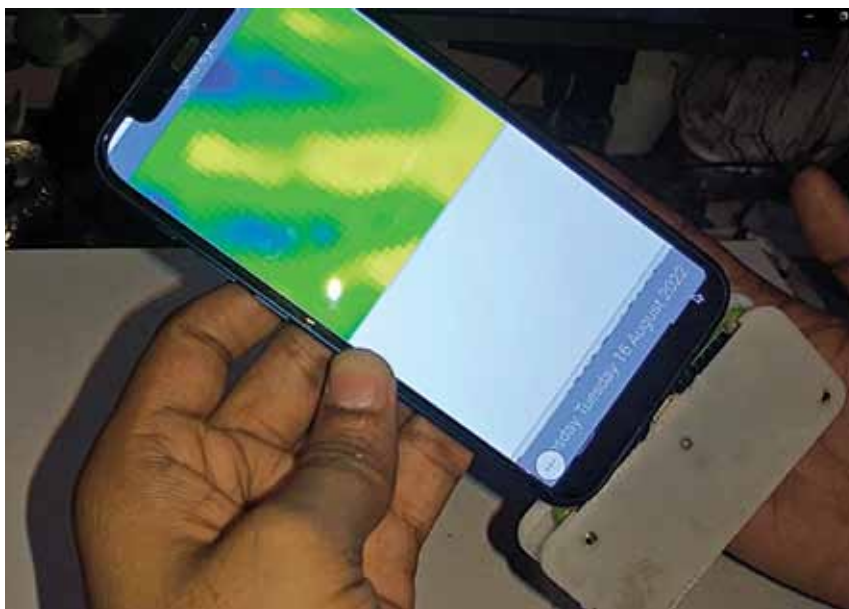


Fig. 2: Prototype attached to a phone's USB port

BILL OF MATERIAL

Components	Quantity	Description
AMG 8833 (S1)	1	8×8 Thermal Sensor
Raspberry Pi Zero W (MCU1)	1	For Processing Sensor Data
Micro USB to iPhone OTG (USB 1)	1	Male iPhone/USB type C to Male micro USB OTG
3D Case	1	PLA/ABS 3D Printed Enclosure